

Dr. Parth Nayak

Data Scientist | AI Researcher | Astrophysicist [he/him]

📍 Heinrich-Wieland-Str. 178, 81735 Munich (Germany)

✉ parth.nayak@physik.lmu.de | 🌐 par-nay.github.io | in dr-parth-nayak-astro | 📷 par-nay



EDUCATION

Ph.D. in Astrophysics (Dr. rer. nat.)

LMU Munich, Germany

Oct 2021 - Dec 2025

- Thesis: Artificial intelligence for harvesting the untapped cosmological information in the Lyman- α forest

B.Sc. + M.Sc. (Dual Degree) in Physics

IISER Bhopal, India

Aug 2015 - Jul 2020

- Thesis: Genetic algorithm for the reconstruction of large angular scale features of cosmic microwave background (CMB)

Higher-secondary School Certificate (HSC)

GSHSEB, India

May 2013 - Mar 2015

- Abitur- / A levels-equivalent; university entrance qualification; subjects: physics, math, chemistry, computer science, English

SKILLS

AI / statistics	Deep learning (CNNs, MLPs, Autoencoders, GANs), simulation-based inference (SBI), genetic algorithms (GA), Gaussian processes (GP), Monte Carlo methods, density estimation
Programming	Python, LaTeX (proficient); HTML, CSS, Javascript (intermediate); C++, Java (basic)
Python packages	NumPy, SciPy, Pandas, Matplotlib, TensorFlow, Keras, Optuna, H5Py, AstroPy, HealPy, <i>among others</i>
OS / Shell	bash, Z shell (proficient), PowerShell (basic)
Other IT skills	High-performance computing (HPC), Git version control, LLM prompt engineering, Markdown, MS / Mac Office apps
Languages	English (full professional proficiency); Gujarati, Hindi (native/bilingual); German (B1/intermediate)

PROFESSIONAL EXPERIENCE

Doctoral Candidate: Astrophysics & Data Science

LMU Munich, Germany

Oct 2021 - Dec 2025

- Conducted research at the **Hightech Agenda Bayern Chair of Astrophysics, Cosmology & AI (ACAI)** on AI applications in cosmology
- Developed a SBI framework “**LyaNNA**” (📄 par-nay/lyanna) based on a novel 1D ResNet with Python/Keras from scratch
 - For complex, realistic, noisy spectroscopic data, **LyaNNA outperforms** conventional methods by **>2x** in inference precision
- Developed “**SynTH**” (📄 par-nay/synth) to model intergalactic absorption from massive cosmological **hydrodynamic simulations** and synthesize training data for **LyaNNA**
- Led two and coauthored one research papers on AI for inference in astrophysics [✍ Nayak+ [2024](#), [2025](#), Chang+ [2025](#)]
- Delivered **10+ technical talks** at **international conferences** and **research institutes** (incl. the Royal Astronomical Society)
- Co-supervised two LMU Master's theses and supervised **5+** numerical and instrumental **lab** courses

Master's Thesis Student: Astrophysics & Data Science

IISER Bhopal, India

Aug 2019 - Jul 2020

- Developed a GA framework “**GA-ILC**” to optimize the reconstruction of CMB from contaminated observations
 - A novel evolutionary algorithm (📄 par-nay/genepy) to extract faint cosmological signals from high-noise environments (foreground removal)
- Published one research paper on evolutionary algorithms in astrophysics [✍ Nayak & Saha [2022](#)]

Summer Intern: Quantum Optomechanics of GW Detectors

MPI Hannover, Germany

May - Jul 2019

- Incorporated modern algorithms for estimating **quantum noise** in gravitational wave (GW) interferometers into **PyGWINC**

Summer Intern: Exoplanet Search

Physical Research Lab, India

May - Jul 2018

- Wrote an order-extraction pipeline for **spectroscopic data reduction** in Python for the PRL Advanced Radial-velocity All-sky Search (**PARAS**), a spectrograph that facilitated India's first exoplanet discovery

Summer Intern: Statistics & Time Series in Astronomy

ARIES Nainital, India May - Jun 2017

- Developed a Python code for **novel statistical tests** of time-variability of active galactic nuclei (AGN) from **noisy photometric data**
 - Contributed to modernizing the **photometric data reduction** pipeline used for observations using a 1.3m telescope in the Himalayas

PUBLICATIONS

Nayak, Walther & Gruen (2025). *LyaNNA II: Field-level inference with noisy Lya forest spectra*. [arXiv:2510.19899](#) (in review)

Chang, Nayak, Walther & Gruen (2025). *Human vs. machine - 1:3. Joint analysis of classical and ML-based summary statistics of the Lyman- α forest*. [arXiv:2508.03264](#) (in review)

Nayak, Walther, Gruen & Adiraju (2024). *LyaNNA: A deep learning field-level inference machine for the Lyman- α forest*. *Astronomy & Astrophysics*, [arXiv:2311.02167](#)

Nayak and Saha (2022). *Application of genetic algorithm to estimate the large angular scale features of cosmic microwave background*. *Monthly Notices of the Royal Astronomical Society*, [arXiv:2102.06569](#)

AWARDS & FELLOWSHIPS**DAAD Research Grant — Ph.D. Scholarship**

Germany Oct 2021 - Dec 2025

By the Deutscher Akademischer Austauschdienst (DAAD) for carrying out frontier research leading to a doctorate in Germany

Proficiency Medal in Physics

India Jul 2020

By the Indian Institute of Science Education and Research (IISER) Bhopal, for the best overall academic performance in the 5-year BS-MS program

DAAD-WISE Scholarship

Germany May - Jul 2019

By the Deutscher Akademischer Austauschdienst (DAAD) for a summer internship at a leading German research institution (MPI Hannover)

Summer Research Fellowship

India May - Jun 2017

By India's Science Academies (IAS, INSA, NASI) jointly, for a summer internship at a leading Indian research institution (ARIES Nainital)

INSPIRE Scholarship for Higher Education

India Aug 2015 - Jul 2020

By the Department of Science & Technology of the Government of India, for pursuing higher education at a premier Indian institution of research & education (IISER Bhopal)

Academic Excellence Award (District Rank 1)

India May 2015

By Shree Vasishtha Vidhyalaya, for achieving the 99.97th percentile state-wide and Rank 1 in the Surat district in the HSC graduation exams

LEADERSHIP & VOLUNTEERING**Co-organizer of "Astronomy on Tap" (AoT) Munich**

Oct 2025 - now

- Coordinating the monthly outreach event to engage the general public with accessible astronomy talks and pub-quizzes

Volunteer at the Long Night of Astronomy

Nov 2024, 2025

- Facilitated science communication and public engagement at the Munich hub (ESO Supernova, Garching) of a major nation-wide event

Co-organizer of "Meet the Speaker" @LMU Observatory

Oct 2023- May 2025

- Revitalized the weekly series by relocating it to the historic 200-year old telescope dome, a unique atmosphere for students for informal networking with international experts

Coordinator & Senior Advisor of IISER Bhopal Astronomy Club (IBAC)

Aug 2016 - Jul 2020

- Played a foundational role as the club's second coordinator, establishing enduring traditions and growing the community into one of India's most popular student astronomy clubs

Physics Department Representative

Aug 2017 - Jul 2018

- Unanimously elected to serve as the primary liaison between faculty and a class of 60+ physics major students

Co-organizer of "Singularity" & TEDxIISER Bhopal

Sep 2017

- Spearheaded external outreach, partnering with local schools to drive student participation in the annual science festival of IISER Bhopal